Lab03 - ME315

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Questão 1

```r
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

Questão 2

```r
fname <- "WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

Questão 3

```r
countries <- read_excel(fname, sheet = "Country")
countries
```

```
# A tibble: 264 x 31
   `Country Code` `Short Name`           `Table Name` `Long Name` `2-alpha code`
   <chr>          <chr>                  <chr>        <chr>       <chr>
 1 ABW            Aruba                  Aruba        Aruba       AW
 2 AFE            Africa Eastern and So~ Africa East~ Africa Eas~ ZH
 3 AFG            Afghanistan            Afghanistan  Islamic St~ AF
 4 AFW            Africa Western and Ce~ Africa West~ Africa Wes~ ZI
 5 AGO            Angola                 Angola       People's R~ AO
 6 ALB            Albania                Albania      Republic o~ AL
 7 AND            Andorra                Andorra      Principali~ AD
 8 ARB            Arab World             Arab World   Arab World  1A
 9 ARE            United Arab Emirates   United Arab~ United Ara~ AE
10 ARG            Argentina              Argentina    Argentine ~ AR
# i 254 more rows
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

Questão 4

```r
countries <- countries %>%
  filter(`Short Name` %in% c("Brazil", "Monaco", "Morocco", "Germany", "Qatar"))
countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name` `Table Name` `Long Name`             `2-alpha code`
  <chr>          <chr>        <chr>        <chr>                   <chr>
1 BRA            Brazil       Brazil       Federative Republic o~ BR
2 DEU            Germany      Germany      Federal Republic of G~ DE
3 MAR            Morocco      Morocco      Kingdom of Morocco      MA
4 MCO            Monaco       Monaco       Principality of Monaco MC
5 QAT            Qatar        Qatar        State of Qatar          QA
```

```
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

Questão 5

```
WDI <- read_excel(fname, sheet = "Data")
```

Questão 6

```
WDI <- WDI %>%
select("Country Code","Indicator Code", "1960":"2018")
```

Questão 7

```
WDI <- WDI %>%
  pivot_longer(
    cols = "1960":"2018",
    names_to = "YEAR",
    values_to = "Valor"
  )
WDI
```

```
# A tibble: 23,421,230 x 4
   `Country Code` `Indicator Code` YEAR  Valor
   <chr>          <chr>            <chr> <dbl>
 1 AFE            EG.CFT.ACCS.ZS   1960     NA
 2 AFE            EG.CFT.ACCS.ZS   1961     NA
 3 AFE            EG.CFT.ACCS.ZS   1962     NA
 4 AFE            EG.CFT.ACCS.ZS   1963     NA
```

```
 5 AFE            EG.CFT.ACCS.ZS   1964      NA
 6 AFE            EG.CFT.ACCS.ZS   1965      NA
 7 AFE            EG.CFT.ACCS.ZS   1966      NA
 8 AFE            EG.CFT.ACCS.ZS   1967      NA
 9 AFE            EG.CFT.ACCS.ZS   1968      NA
10 AFE            EG.CFT.ACCS.ZS   1969      NA
# i 23,421,220 more rows
```

Questão 8

```
WDI <- WDI|>
  mutate(YEAR = as.integer(YEAR))
```

Questão 9

```
WDI_final <- WDI %>%
  semi_join(countries, by = "Country Code") %>%
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") %>%
  left_join(countries %>%
      select(`Country Code`, `Short Name`), by = "Country Code") %>%
  pivot_wider(names_from = `Indicator Code`, values_from = Valor) %>%
  select(`Country Code`, `Short Name`, YEAR, `SP.DYN.CBRT.IN`)

WDI_final
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
   <chr>          <chr>        <int>          <dbl>
 1 BRA            Brazil        1960           43.8
 2 BRA            Brazil        1961           43.3
 3 BRA            Brazil        1962           42.7
 4 BRA            Brazil        1963           42.0
 5 BRA            Brazil        1964           41.0
 6 BRA            Brazil        1965           39.8
 7 BRA            Brazil        1966           38.6
 8 BRA            Brazil        1967           37.4
 9 BRA            Brazil        1968           36.3
```

```
10 BRA             Brazil        1969            35.7
# i 285 more rows
```

Questão 10

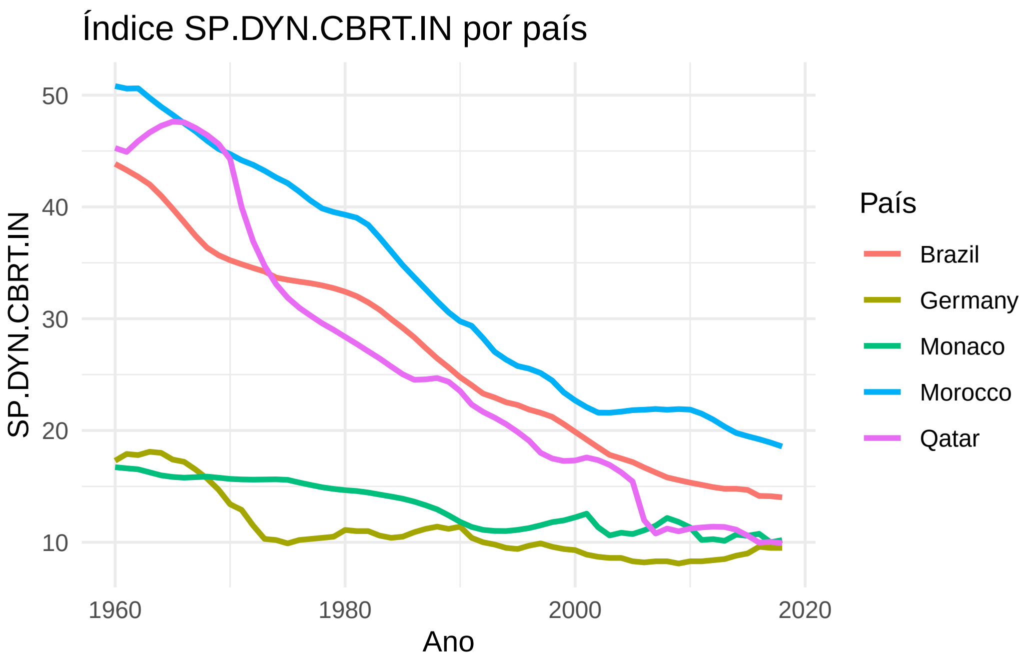
```r
WDI_final <- WDI_final %>%
  arrange(`Short Name`, YEAR)

WDI_final
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
   <chr>          <chr>        <int>          <dbl>
 1 BRA            Brazil        1960           43.8
 2 BRA            Brazil        1961           43.3
 3 BRA            Brazil        1962           42.7
 4 BRA            Brazil        1963           42.0
 5 BRA            Brazil        1964           41.0
 6 BRA            Brazil        1965           39.8
 7 BRA            Brazil        1966           38.6
 8 BRA            Brazil        1967           37.4
 9 BRA            Brazil        1968           36.3
10 BRA            Brazil        1969           35.7
# i 285 more rows
```

Questão 11

```r
ggplot(WDI_final, aes(x = YEAR, y = `SP.DYN.CBRT.IN`, color = `Short Name`)) +
  geom_line(linewidth = 1) +
  labs(title = "Índice SP.DYN.CBRT.IN por país",
       x = "Ano",
       y = "SP.DYN.CBRT.IN",
       color = "País") +
  theme_minimal()
```




```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-05 12:12:24 -03"
```